

A Physics-Aware Benchmark for Phase-Resolved Neural Operator Learning in Boiling Flows: Statistically Rigorous Evidence for Condition-Dependent Architecture Trade-offs

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Abstract

Neural surrogates for boiling flows must forecast smooth thermal structure, sharp liquid-vapor interfaces, and locally conservative transport over long autoregressive horizons. We introduce a reproducible, statistically rigorous benchmark for this task and use it to show that architecture rankings for phase-resolved boiling forecasting are highly condition-dependent — a finding with direct implications for how the field should evaluate operator-learning claims going forward. On a fixed tutorial-scale trajectory split, a Tucker-factorized Fourier Neural Operator (T-FNO) and a parameter-comparable U-Net exhibit a statistically significant Pareto trade-off (paired bootstrap intervals, exact sign-flip tests, Holm-Bonferroni correction, $n=11$ seeds): T-FNO is significantly better on interface-temperature fidelity, U-Net is significantly better on mass-conservation error. When this same comparison is repeated on two independently held-out physical conditions from the same source dataset, however, the trade-off does not reproduce: U-Net is descriptively favored over T-FNO on every primary metric, including interface temperature, though the five-seed independent test is statistically underpowered. A predeclared spectral divergence-penalty loss, applied to a local-global hybrid architecture, restores mass-conservation non-inferiority to U-Net on the original split ($p=.000488$) and transfers the conservation advantage directionally to the

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independent conditions, though not yet with confirmatory statistical power there. We additionally identify and correct a previously unreported evaluation pitfall — phase-indicator (vapor-fraction) predictions exceeding their physical $[0,1]$ range during autoregressive rollout, which we show can materially confound downstream interface and dry-area diagnostics — and demonstrate a predeclared statistical decision-gate methodology that correctly rejects a promising but non-replicating single-seed observation. A critical-heat-flux (CHF)-motivated heater-adjacent dry-area proxy is evaluated across all trajectories but reveals no sustained ground-truth event in any of them, so no claim of validated CHF detection is made. We contribute the benchmark protocol, the physical-validity check, the decision-gate methodology, and — as the central empirical finding — direct evidence that single-condition architecture comparisons in this domain risk reporting a property of the test trajectory rather than a genuine architectural property, with a concrete demonstration of how to detect this via independent-condition replication.

Keywords: neural operator, Fourier neural operator, pool boiling, phase-resolved forecasting, conservation laws, BubbleML

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1. Introduction

Pool boiling is central to high-power electronics cooling, data-center thermal management, and nuclear thermal-hydraulics, where heat fluxes exceed single-phase capabilities. The mechanism's effectiveness is bounded by the critical heat flux (CHF)—the point at which a growing vapor film insulates the

heated surface, triggering runaway temperature excursions. Classical correlations (e.g., Zuber's hydrodynamic model) estimate aggregate CHF thresholds but do not resolve transient local bubble dynamics. While deployment-grade CHF forecasting requires synchronized wall-heat-flux measurements and verified boiling-crisis transitions—neither of which is present in available open simulation benchmarks—resolving localized, heater-adjacent dry-area dynamics serves as a diagnostic for phase-interface integrity. In this work, we evaluate a heater-adjacent dry-area fraction proxy across stable boiling trajectories strictly as an autoregressive physical-stability test for neural operator rollouts, explicitly distinguishing phase-validity stress-testing from validated CHF event detection.

High-fidelity interface-tracking computational fluid dynamics (CFD) can resolve bubble nucleation, growth, coalescence, and departure, but its cost limits repeated design exploration and real-time digital-twin use. Operator learning offers a route to fast state-to-state surrogates learned directly from CFD trajectories. The Fourier Neural Operator (FNO) is an attractive candidate because it represents global spatial coupling through a resolution-invariant spectral representation. Boiling fields, however, combine non-periodic domain boundaries with a sharp, near-discontinuous liquid-vapor interface — properties in tension with FNO's implicit periodicity assumption and its low-frequency spectral truncation. The BubbleML benchmark reported that a vanilla FNO underperforms a U-Net on pool-boiling temperature forecasting by roughly a factor of two in boundary RMSE (not the order-of-magnitude gap sometimes assumed in early framing of this problem), attributed tentatively to convolutional edge sensitivity versus the Fourier layer's periodicity assumption.

This motivates a natural question: do parameter-efficient FNO variants close this gap on the full, phase-resolved multi-field boiling state (not just temperature), and — critically — **is any observed architecture ranking a genuine, generalizable property of the architectures, or an artifact of the specific trajectory it was measured on?** The second half of this question is, to our knowledge, not systematically addressed in prior boiling-operator benchmark work, which typically reports single-trajectory or single-seed comparisons. We treat it as a first-class experimental question rather than an assumed non-issue.

This paper makes five contributions:

1. A reproducible, physics-aware benchmark protocol for five-field (velocity,

pressure gradient, temperature, vapor-phase indicator) phase-resolved boiling forecasting, evaluated with paired bootstrap intervals, exact sign-flip tests, and Holm-Bonferroni correction across multiple seeds.

2. Identification and correction of a previously unreported evaluation pitfall: phase-indicator predictions exceeding their physical $[0,1]$ range under autoregressive rollout, and a demonstration that this materially confounds downstream evaluation metrics that depend on the phase field.
3. A predeclared statistical decision-gate methodology, demonstrated concretely by its correct rejection of a promising but non-replicating single-seed rollout observation — a discipline we argue should be standard practice before committing engineering effort to an observed effect.
4. **A cross-condition replication check**, in which a statistically confirmed architecture trade-off (T-FNO favors interface fidelity, U-Net favors mass conservation) on one trajectory split is re-tested under an independent protocol on two held-out conditions from the same source dataset — and does not reproduce. The protocol also differs in spatial resolution, so this is evidence against generalizing the tutorial ranking, not evidence that wall temperature alone caused the ranking change.
5. A predeclared, mechanistically motivated conservation intervention (a spectral velocity-divergence penalty) that restores mass-conservation non-inferiority to U-Net on the original split and transfers the conservation advantage directionally — though not yet confirmatorily — to the independent conditions, illustrating how a targeted physical constraint, rather than architecture complexity alone, is needed to move a specific objective.

Consistent with contribution 4, our central finding is not a single winning architecture. The tutorial ranking did not replicate under an independent cross-condition protocol that also differed in spatial resolution. Reporting a ranking from a single trajectory—however statistically rigorous the seed-level inference—is therefore not sufficient evidence of a generalizable architectural property. This is a methodological finding of independent value to the broader neural-operator-for-science community, beyond its specific application to boiling.

2. Related Work

2.1 Classical CHF prediction and boiling simulation

Critical heat flux has been studied mechanistically since the mid-twentieth century, most influentially through Zuber's hydrodynamic instability model[?]. Engineering practice also uses application-specific correlations for CHF estimation. These are computationally simple aggregate predictors, but they do not resolve transient local bubble dynamics on a specific surface and offer no natural mechanism for real-time, sensor-driven early warning.

Direct numerical resolution of boiling requires interface-tracking methods — volume-of-fluid, level-set, or phase-field formulations coupled to the Navier-Stokes and energy equations. The BubbleML dataset[?], which underlies the present work's data, was generated with the Flash-X interface-tracking solver and spans multiple fluids, heater geometries, and boiling regimes. Its accompanying benchmark paper is, to our knowledge, the first systematic comparison of neural operator architectures against convolutional baselines on boiling data, and its reported $\sim 2\times$ FNO-versus-U-Net temperature BRMSE gap is the empirical starting point for the present investigation.

2.2 Operator learning and convolutional surrogates

The Fourier Neural Operator[?] performs global convolution via pointwise multiplication in a truncated Fourier domain, and has demonstrated strong performance on smooth, largely periodic PDE systems. Convolutional U-Net architectures[?] instead rely on local receptive fields and multi-scale skip connections, and serve as a strong general-purpose baseline for dense spatial prediction, including PDE surrogate modeling.

Several parameter-efficient FNO variants address the capacity and generalization trade-offs of the original architecture. Tensor-factorized (Tucker) FNO represents the spectral weight tensor in a low-rank Tucker decomposition; axis-factorized FNO (F-FNO)[?] factorizes the spectral convolution along each spatial axis independently. The BubbleML benchmark's own results indicate both variants outperform vanilla FNO on boiling forecasting despite having substantially fewer parameters, attributed to reduced overfitting to excess spectral capacity rather than a fix to the underlying periodicity or discontinuity issue. U-shaped FNO (UNO)[?] grafts U-Net-style multi-scale structure onto Fourier blocks; the LOGLO-FNO family of Kalimuthu et al. (2025)[?] explicitly combines local and global spectral branches to improve local, high-frequency representation. Transformer-based operators, including